Report R5 — Dissipative rules: recurrent classes, attractors, protected coherence, and growth laws

Krylov sector analysis of quantum cellular automata (Tier 1a/1b/1c)

QCA Sector Analysis project (Tier 1)

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Abstract

We extend the exact pipeline to the 240 dissipative quantum cellular automata (rules containing a reset channel `D` or `E`). The one-cycle channel is now genuinely non-unitary: the directed transition graph has *transient* strongly connected components flowing into *recurrent classes*. We sweep all dissipative rules (both boundary conventions, $N = 6 \dots 13$), census their attractor structure, classify each recurrent class by the peripheral spectrum of its exact restricted superoperator (Tier 1b), and — new in this revision — extract the *growth laws* of the attractor count and the largest attractor (Tier 1c). Two results organise the report. First, the transition graph is not a lossy picture of the channel but the *exact* description of a different experiment (the monitored QCA), and it fails to describe the unmonitored channel in a way we can now census exactly: for 92 of 240 `pbc` rules the channel's attractor is not spanned by computational-basis states at all, and every one of those rules contains a Hadamard, while all 80 Hadamard-free dissipative rules are basis-spanned without exception. Second, dissipative growth is far from generic: 58 of the `pbc` series satisfy an *exact* integer linear recurrence, so their growth bases are algebraic numbers — φ, $\sqrt{3}$, the supergolden and plastic constants — rather than fitted slopes.

1 Dissipative graph structure

For a dissipative rule the channel $\Phi_C(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger$ has ≥ 2 Kraus operators. The exact engine propagates every Kraus branch (never merging incoherent alternatives) and records the directed transition graph $M_{yx} = \sum_\mu |\langle y|K_\mu|x\rangle|^2$. Its terminal strongly connected components are the recurrent classes; non-terminal components are transient. We record, per $(\text{rule}, N, \text{bc})$: the number and sizes of recurrent classes, the transient depth (longest condensation-DAG path), and the basin structure. The engine and its dissipative branch machinery are validated against an independent dense-channel oracle (528/528 support checks; per-site Kraus completeness $A_0^\dagger A_0 + A_1^\dagger A_1 = \mathbb{1}$).

1.1 What actually fires each cycle: neighbour-controlled Kraus operators

Saying that "the reset channels fire every cycle" is true only in the weak sense that every site is *addressed* every cycle; whether anything happens to it is decided by its two neighbours. Within one brickwork sublayer the sites being acted on are mutually non-adjacent, so the neighbours of an active site lie in the *other* sublayer and are untouched while it acts. The sublayer is therefore a genuinely *controlled* channel: writing $P_i^{(mn)} = |m\rangle\langle m|_{i-1} \otimes |n\rangle\langle n|_{i+1}$ for the projector on the neighbour pair, the site-i Kraus operators are

$$K_\mu^{(i)} = \sum_{m,n\in\{0,1\}} P_i^{(mn)} \, c_\mu^{(mn)} \, P_i^{(mn)}, \tag{1}$$

with the local symbol $r_{mn} \in \{\mathtt{I}, \mathtt{V}, \mathtt{D}, \mathtt{E}\}$ fixing the branches

$$\begin{array}{ll} \texttt{I:} & c_0 = \mathbb{1} \\ \texttt{V:} & c_0 = H \\ \texttt{D:} & c_0 = |0\rangle\langle 0|, \quad c_1 = |0\rangle\langle 1| \\ \texttt{E:} & c_0 = |1\rangle\langle 1|, \quad c_1 = |1\rangle\langle 0| \end{array}$$

(the two branches of a reset record whether the centre had to be de-excited). Each summand of (1) literally reads *projector–decay–projector*: the neighbour projector selects the configuration, the local map acts, and the projector still holds afterwards because $P_i^{(mn)}$ commutes with everything supported on site i alone. That commutation is what makes the control *classical within a sublayer* — the conditioning register is never disturbed by the operation it conditions — and it is the reason the Kraus branch label μ can be carried around as a per-site classical record at all. The practical consequence is that dissipation here is not a rate: a rule with $r_{00} = \texttt{D}$ and `I` elsewhere acts as the identity on any configuration containing no 000 pattern, so which rules are strongly damped is a statement about the *local patterns* the state contains, not about a coupling constant.

What M is, exactly. It is worth being precise about the status of M, because the usual phrase "the graph is an approximation" is wrong in a way that matters. M is the transition matrix of the QCA *monitored in the computational basis after every cycle*: measure all N qubits at the end of each brickwork cycle and M gives the exact conditional distribution of outcomes. So M is not an approximation of anything — it is the exact dynamics of a different (and physically realisable) experiment. What it cannot do is predict the *unmonitored* channel, because there coherences feed back into populations between cycles. Everything below is a statement about how far apart those two experiments are, rule by rule.

Individual trajectories versus the graph. The distinction is easiest to state at the level of single trajectories, and it is not the usual "trajectories average to the channel" remark. A trajectory of the *unmonitored* channel is a sequence of Kraus labels μ, one per site per cycle. Between resets the conditional state is *not* a basis state: `V` is a Hadamard, so a trajectory generically carries superpositions, and the operators applied later in the same cycle are controlled by neighbours that are themselves in superposition. Equation (1) then acts as a *coherently controlled* gate — a partial swap of amplitude between the neighbour-configuration branches — rather than as a classical if-then. The branch probability of the next label depends on the accumulated amplitudes, i.e. on the whole measurement history, so an unmonitored trajectory is a walk on a continuum of rays, not on 2^N labels. (This is exactly why the "ray graph" repair fails: rules 22 and 50 already exceed 6×10^4 distinct rays at $N = 3$, where the basis graph has 8 nodes.)

In the *monitored* experiment all N qubits are measured at the end of each cycle. The state is projected onto a basis label, so the next cycle's branch probabilities depend on that label alone and nothing else — which is precisely the Markov property, with kernel M_{yx}. Two things are destroyed: coherence generated by `V` never survives into the following cycle, and the trajectory ensemble is re-collapsed to populations once per cycle. Note that M is *not* the "dephase after every gate" chain: within a cycle the Hadamards and the controlled resets act coherently, and only the end-of-cycle measurement intervenes. Consequently M reproduces the unmonitored channel's populations exactly iff no coherence created in one cycle can influence populations in the next, which happens precisely when every Kraus operator is a 0/1 partial permutation — i.e. when the rule is Hadamard-free. That criterion is confirmed without exception in §5: all 80 `V`-free dissipative rules are basis-spanned, and every rule that is not contains a `V`.

2 Census of attractor structure

Table 1 classifies every dissipative rule by the structure of M at the largest computed N: a unique classical fixed point, several point attractors (multistability, signalling a basis-diagonal conserved charge), or one or more extended (>1-state) attractors.

attractor structure	obc0	pbc
unique fixed point	66	36
multistable (point attractors)	77	75
single extended attractor	63	47
multiple extended attractors	34	82
total dissipative rules	240	240

Table 1: Dissipative rules by attractor structure of the monitored chain M and boundary convention.

Two ground-truth items (context §8) are reproduced: rule 22 (`IVVD`) on `pbc` at even N has exactly three point attractors (the vacuum and the two Néel states); rules 28/29 illustrate the boundary/wall-transparency contrast.

3 Tier 1b: attractor types

For each recurrent class R we build the *exact* restricted superoperator Φ_R on $\mathrm{span}(R)\otimes\mathrm{span}(R)^*$ from the labelled Kraus branches, and read its peripheral spectrum (geometric multiplicities, robust to the non-normal transient Jordan structure). A class is *mixing* (a unique steady state; a classical pointer/dark state when $|R| = 1$), a *limit cycle* (peripheral roots of unity, basis-diagonal eigenoperators), or *coherence-carrying* (a decoherence-free subsystem of dimension $d_k > 1$).

W	tuple	#recur	attractor kinds	max d_k
0	`DDDD`	1	1×mixing	1
1	`VDDD`	1	1×unresolved	1
22	`IVVD`	3	3×mixing	1
28	`IIVD`	11	3×mixing, 8×coherent	3
29	`EIVD`	10	2×mixing, 8×coherent	3
50	`DVVV`	1	1×mixing	1
76	`IIID`	131	131×mixing	1
178	`DVVE`	2	2×mixing	1
200	`DIII`	90	90×mixing	1
232	`DIIE`	46	46×mixing	1

Table 2: Tier-1b attractor classification (pbc, $N = 8$). Rule 28 carries coherence-protecting extended attractors ($d_k > 1$); most others relax to classical pointer states.

4 Strong vs. weak symmetry: what each stage can see

A *strong* symmetry commutes with every Kraus operator individually; a *weak* symmetry only makes the channel covariant (context §5). For unitary rules there is a single Kraus operator and the two coincide. This pipeline does **not** detect symmetry *operators* (no commutant of $\{K_\mu\}$, no charge null-space — that is Tier 2); it separates the two *operationally*:

- **Tier 1a (graph).** M is the monitored chain. Only basis-diagonal *strong* charges fragment it; their signature is the number of recurrent classes (Table 1). This is the only symmetry information the graph carries.

- **Tier 1b (channel).** The exact Cesaro rank $\dim \mathrm{Fix}(\Phi)$ (geometric multiplicity of eigenvalue 1 of the full 4^N superoperator) sees the coherent fixed-point algebra. Weak symmetries and protected coherences appear here and *only* here.

The gap $\dim \mathrm{Fix}(\Phi) - \sum_R \dim \mathrm{Fix}(\Phi_R) \geq 0$ measures what the monitored description misses (Table 4).

5 When the graph cannot name the attractor: two distinct failure modes

The earlier revision of this report reported a single number — "38 of 240 rules are unfaithful" — from a leakage test. That test is the weaker of two, and it misses the more common failure. We now separate them.

Mode 1: leakage (the weak test). Ask whether $\Phi^t(\mathbb{1}/d)$ retains population on states the graph calls transient, and whether $\mathrm{Fix}(\Phi)$ has weight outside the recurrent block. This catches rules where the graph puts the attractor on the *wrong states*: **38** of 240 at $N = 4$, `pbc` (Table 3). Rule 22 is the sharp case — fixed-point operators carry weight 0.49 outside the recurrent block and $\Phi^{400}(\mathbb{1}/d)$ retains 12.5% of its population on graph-"transient" states (limit support: 11 basis states, not 3). Since $\dim \mathrm{Fix} = 25 = 5^2$ the true attractor is a 5-dimensional decoherence-free subspace, and the "three point attractors" are an artifact of dephasing. Rule 50 ($\dim \mathrm{Fix} = 13$, 25% leaked) and rule 106 (42%) are stronger cases.

Mode 2: coherence (the structural test). A rule can pass the leakage test and still be undescribable by any state-space graph, because the graph's vertices *are* basis states: whatever it calls the attractor is a *set*, while the channel's attractor is a *subspace*. The right question is therefore whether that subspace is basis-spanned,

$$\dim_{\text{att}} = \dim \operatorname{supp} \lim_{T\to\infty} \tfrac{1}{T} \sum_{n<T} \Phi^n(\mathbb{1}/d), \qquad \text{coh dirs} = \dim_{\text{att}} - \#\{\text{basis states inside}\},$$

and coh dirs > 0 is exactly the structure no graph can represent. Rule 27 (`VEVD`) shows the two tests are genuinely different: the graph reports 9 recurrent states, the truth is *one* pure stationary superposition — and the leak is 0, so Mode 1 sees nothing. Unlike the leaked weight, coh dirs is initial-state independent and not normalised by 2^N (leakage from $\mathbb{1}/d$ gives rules 104/233 a spurious $2/2^N$ that vanishes with N for purely dilutional reasons).

The Cesaro limit must be taken to a genuine *spectral gap*, not to a fixed burn-in. With burn-in 600, rule 19 at $N = 6$ reports $\dim_{\text{att}} = 35$ with smallest kept eigenvalue $1.9 \cdot 10^{-9}$ against largest dropped $4.8 \cdot 10^{-10}$: no gap, so the "attractor" is mostly still-decaying transient. Converged (burn-in 4000) it is 5, with a gap of $1.8 \cdot 10^{-2}$ against 10^{-15}. We therefore double the burn-in until $\lambda_{\text{kept}}/\lambda_{\text{dropped}} > 10^4$.

Census. All 240 dissipative rules, `pbc`, $N = 4, 6, 8$, every rule converged:

	$N=4$	$N=6$	$N=8$
rules with a non-basis-spanned attractor	84	60	72
not converged	0	0	0

The count is **non-monotonic** in N and stays extensive: 54 rules are coherent at all three sizes and 92 at some size, so this is not a small-N accident that washes out. Two structural facts stand out:

- **The Hadamard is necessary.** All 80 Hadamard-free dissipative rules (`V` $\notin$ tuple) are basis-spanned at every censused size, and all 92 coherent rules contain at least one `V`. With only `I`, `D`, `E` the Kraus operators are basis permutations composed with resets, so the channel maps basis-diagonal states to basis-diagonal states and the graph is complete by construction.

- **The wall-Hadamard core.** The 12 rules with $r_{01} = r_{10} =$ `V` — $\{18, 19, 22, 23, 50, 55, 146, 151, 178, 179, 18$ i.e. Hadamard exactly on the domain-wall configurations — are coherent *without exception.* A Hadamard driven by the wall pattern rather than by the site occupation is the mechanism that keeps a superposition stationary.

Mode 2 also cuts across the graph-level classification of §2: 37 of the 47 `pbc` rules with a single extended attractor are coherent, against 13 of 75 multistable rules — extended graph attractors are the ones most likely not to be attractors at all in the coherent sense.

W	tuple	#rec. states	dim Fix(Φ)	leaked weight	graph faithful?
0	`DDDD`	1	1	0.000	yes
200	`DIII`	10	100	0.000	yes
232	`DIIE`	6	36	0.000	yes
28	`IIVD`	3	9	0.000	yes
76	`IIID`	11	121	0.000	yes
22	`IVVD`	3	25	0.125	**no**
50	`DVVV`	1	13	0.250	**no**
178	`DVVE`	2	9	0.063	**no**
106	`DEIV`	1	8	0.422	**no**
census at $N = 4$, pbc: 202 faithful / 38 unfaithful of 240 dissipative rules					

Table 3: Mode-1 (leakage) test (pbc, $N = 4$): leaked weight is the population of $\Phi^{400}(\mathbb{1}/d)$ remaining on graph-transient states. Non-zero leak means the graph misplaces the coherent attractor.

W	tuple	classical dim	Cesaro rank	coherence gap
0	`DDDD`	1	1	0
1	`VDDD`	15	15	0
22	`IVVD`	3	25	22
28	`IIVD`	3	9	6
50	`DVVV`	1	13	12
178	`DVVE`	2	9	7
200	`DIII`	10	100	90
232	`DIIE`	6	36	30

Table 4: Exact Cesaro rank vs. the graph-resolved per-class fixed dimension (pbc, $N = 4$). A zero gap (rule 0, pure decay) is fully classical; a large gap flags protected coherence.

6 Tier 1c: dissipative growth laws

We now fit the N-dependence of the attractor count #att, the largest attractor $D_{\max}$, the largest basin, and the transient depth for all 240 rules and both boundary conventions over $N = 6 \dots 13$

(coverage is complete; no rule is ergodic). Two methodological points were forced by the data and are worth recording, since they are the difference between a result and an artifact.

Parity. Dissipative series oscillate between even and odd N — a `pbc` chain of odd length cannot host the two Néel states — so a joint fit is meaningless. We fit all N, even N, and odd N separately and declare a *parity split* when the split halves the rms ln-residual; 43 `pbc` rules (16 `obc0`) split.

Rate model. The unitary pipeline fits $\ln y = c + \alpha \ln N + \kappa N$, carrying the $\alpha \ln N$ term so that binomial prefactors do not bias κ. On the ragged dissipative series that model is unstable: for rule 90, $D_{\max} = 1, 15, 1, 7, 3, 63, 2, 127$ yields $\alpha = -7.5$ and a growth base of 11.25 — impossible, since $D_{\max} \le 2^N$ forces base ≤ 2. We therefore take the growth *class* from BIC model selection but the growth *rate* from the two-parameter fit $\ln y = c + \kappa N$, which cannot trade the rate against a power, and mark a series *irregular* (no rate reported) if it exceeds the physical bound or keeps an rms residual above 0.35 after the parity split.

Results. For `pbc`: #att is constant for 140 rules, exponential for 72, polynomial for 16, irregular for 12; $D_{\max}$ is constant for 109, exponential for 68, polynomial for 41, irregular for 22. `obc0` is markedly tamer (162 constant, 54 exponential, 22 polynomial, 2 irregular for #att), the same open-vs-periodic asymmetry seen for the unitary rules. The largest rates are $\kappa \approx 0.61$ for #att (rule 76) and $\kappa \approx 0.64$ for $D_{\max}$ (rule 67), both comfortably below $\ln 2$.

W	tuple	#att	$D_{\max}$	#att growth (base)	$D_{\max}$ growth (base)	attractor cla
0	`DDDD`	1	1	constant	constant	unique-point
4	`IDDD`	3571	1	exponential (1.6180, golden φ)	constant	multistable
22	`IVVD`	3	1	constant*	polynomial*	multistable
28	`IIVD`	120	32	exponential (1.3247, plastic ρ)	irregular	multi-extend
50	`DVVV`	1	1	constant	constant	unique-point
146	`DVVI`	2	1	constant	constant	multistable
178	`DVVE`	2	1	constant	constant	multistable
200	`DIII`	14197	1	exponential (1.7549, root of $x^3 = 2x^2 - x + 1$)	constant	multistable
232	`DIIE`	3572	1	exponential (1.6180, golden φ)	constant	multistable
3	`VVDD`	1	6561	constant	exponential (1.7321, $\sqrt{3}$)	extended
19	`VVVD`	1	25889	constant	exponential (1.8872)	extended
35	`VVDV`	1	18464	constant	exponential (1.8478)	extended

Table 5: Growth laws for representative dissipative rules (pbc, $N = 6 \dots 13$). * marks a series whose even- and odd-N subsequences obey different laws. Where an exact integer recurrence exists, the base is the named algebraic constant rather than a fitted slope.

6.1 Exact algebraic growth bases

For 58 of the `pbc` series (and 81 `obc0`) the integers themselves satisfy an *exact* linear recurrence $a(n) = \sum_i c_i \, a(n-i)$ with integer coefficients, solved over $\mathbb{Q}$ on the first equations and then *verified against every remaining term.* For those the growth base is the largest root of the characteristic polynomial: an algebraic number, not a regression estimate. They cluster on a handful of constants (Figure 3):

base	constant	pbc	obc0
1.618034	golden φ	26	33
1.732051	$\sqrt{3}$	10	16
1.465571	supergolden ψ	10	15
1.324718	plastic ρ	8	13
1.839287	tribonacci ψ	2	2
1.754878	root of $x^3 = 2x^2 - x + 1$	2	2

Representative cases, all verified on the full series:

- W4 (`IDDD`): #att $= 18, 29, 47, 76, 123, 199, 322, 521$ obeys $a(n) = a(n-1) + a(n-2)$ — Lucas numbers, base φ.
- W3 (`VVDD`): $D_{\max}$ obeys $a(n) = 3\,a(n-2)$, base $\sqrt{3}$.
- W200 (`DIII`): #att obeys $a(n) = 2a(n-1) - a(n-2) + a(n-3)$, base 1.754878.
- W5 (`EDDD`): #att obeys $a(n) = a(n-2) + a(n-3)$, base plastic ρ.
- W36/44/72/73: #att obeys $a(n) = a(n-1) + a(n-3)$, supergolden.

These are the counting signatures of local constraints: a Fibonacci-type recurrence is what one gets when a configuration is admissible unless it contains a forbidden local pattern, so the attractor count is a transfer-matrix count over a constrained chain and its base is that transfer matrix's Perron root. This is the same mechanism as the unitary sector counts (R2), now visible in the dissipative attractor structure.

6.2 The growth-rate map

Figure 1 places every rule at its pair of exponential rates. Figure 2 is the same map with the structural annotation: hue and marker give the graph-level attractor class, a hollow ring marks a rule whose attractor is *not* basis-spanned (§5, Mode 2), and a purple halo marks a parity split. The map has visible structure rather than a cloud: rules concentrate on the axes ($\kappa \approx 0$ for one quantity while the other grows), the coherent rules sit overwhelmingly in the "extended attractor" band at high $\kappa_{D_{\max}}$, and the multistable rules occupy a separate arm along $\kappa_{D_{\max}} \approx 0$ reaching $\kappa_{\#\mathrm{att}} \approx 0.6$. Irregular rules carry no rate and are absent from both panels (26 of 240 for `pbc`, 5 for `obc0`).

7 Transient relaxation

Figure 4 shows the transient depth (an exact integer proxy for the relaxation time) growing with N for representative multistable rules.

8 Status and next steps

The graph-level dissipative classification, the Tier-1b per-class attractor types, the coherent-attractor census and the Tier-1c growth laws are complete and checkpointed. The open targets are: (i) whether the wall-Hadamard core's coherent attractor is a decoherence-free subsystem with a Tier-2 symmetry operator behind it; (ii) why the coherent count is non-monotonic in N ($84 \to 60 \to 72$), which looks like a parity effect of the same kind as §6 but is not yet proved; and (iii) pushing the census beyond $N = 8$, currently limited by the 2^N dense ρ.

All numbers derive from `results/*.jsonl` (engine_version `1a.1`), `analytics/coherent_attractor_census` and `figures/dissipative_summary.csv`; regenerated by `python -m qca_fragmentation.quantum.dissipat` `python -m qca_fragmentation.quantum.attractor_structure`, `python -m qca_fragmentation.scaling.c` and `python -m qca_fragmentation.scaling.diss_figure`.

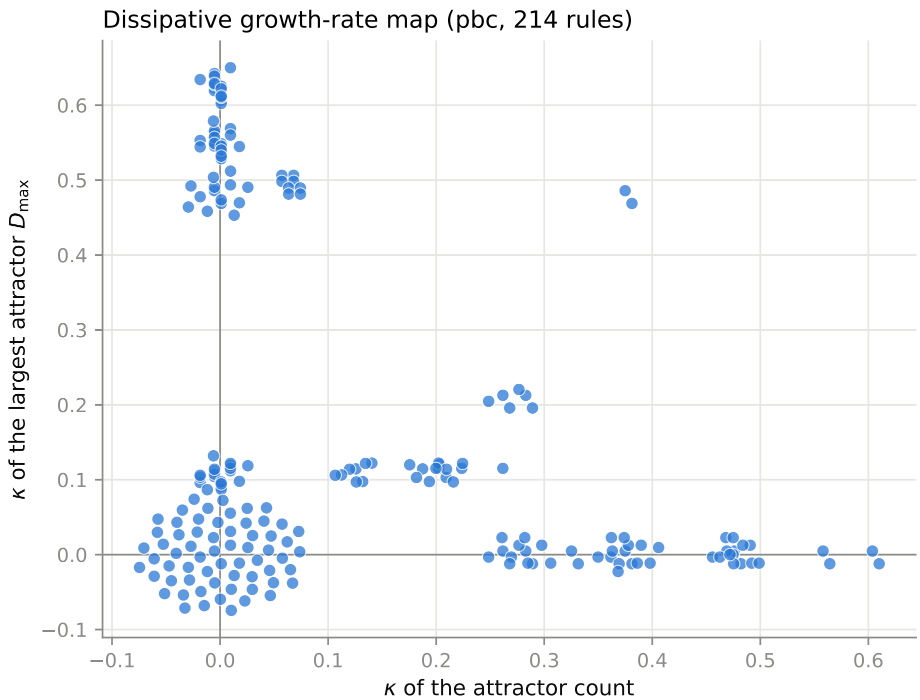

Figure 1: Dissipative growth-rate map (pbc): exponential rate of the attractor count against that of the largest attractor, one point per rule. Coincident points are spread over a small disc so that they can be counted.

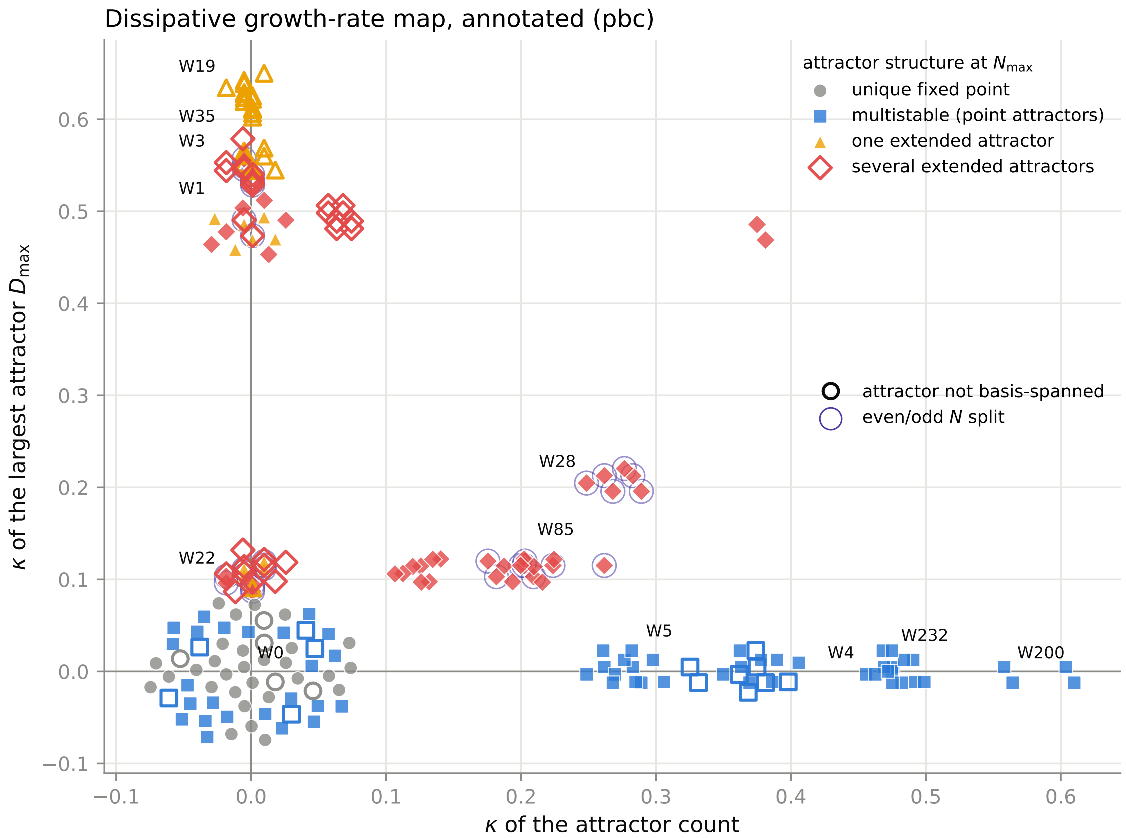

Figure 2: The same map, annotated. Hue/marker: graph-level attractor structure at $N_{\max}$. Hollow ring: the channel attractor is not spanned by basis states, so no state-space graph can name it. Purple halo: even- and odd-N subsequences obey different growth laws.

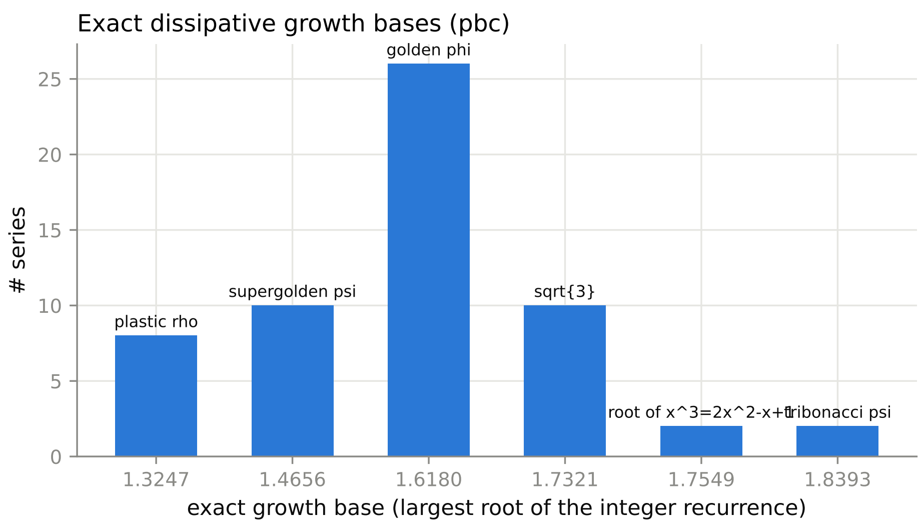

Figure 3: Exact growth bases (pbc): the largest root of each verified integer recurrence. The bases are not spread over an interval — they pile up on a few algebraic constants.

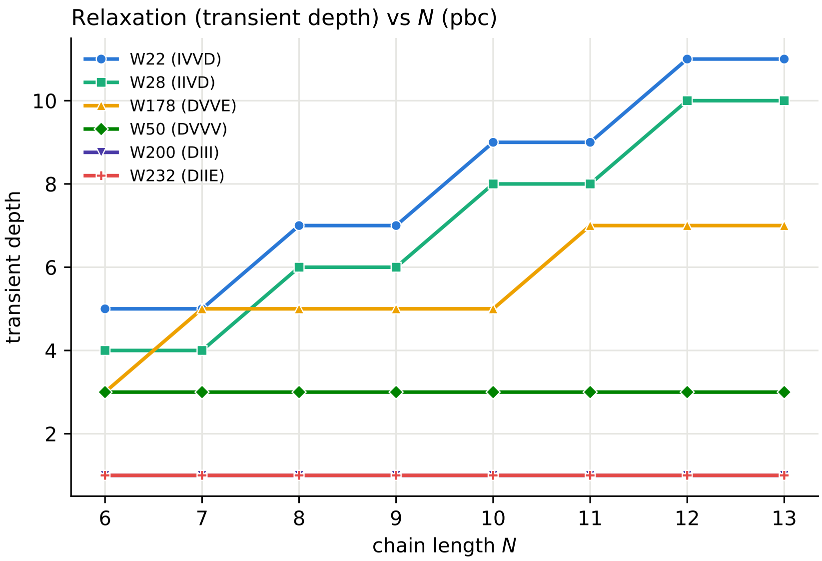

Figure 4: Transient depth vs. N (pbc) for selected dissipative rules.